

1) a) $\Delta H_r^\circ = +69.36 \text{ kJ/mol}$

- Since $\Delta H > 0$, the reaction is endothermic
- To maintain constant temperature at 25°C , heat must be added to the reactor
- If the reactor operates adiabatically ($Q = 0$):
 - The reaction absorbs heat from the system, then temperature will decrease
- Bond energy interpretation:

Energy required to break reactant bonds $>$ Energy released when forming product bonds

b) $\Delta U_r^\circ = \Delta H_r^\circ - \Delta n_g RT$

gas mole: $\Delta n_g = (2+5) - 0 = 7$

Sub: $\Delta U_r^\circ = 69.36 - (7)(8.314 \times 10^{-3})(298)$

$\Rightarrow \Delta U_r^\circ = 52.02 \text{ kJ/mol}$

$\Rightarrow \Delta U > 0 \Rightarrow$ internal energy increases

$\Rightarrow$ Energy must be supplied to drive the reaction

c) $n_{\text{CaCl}_2} = \frac{150.0}{64.10} = 2.34 \text{ mol}$

• rigid $\Rightarrow w = 0$

• Closed system: $Q = \Delta U$

$Q = n \cdot \Delta U_r^\circ$

$\Rightarrow Q = (2.34)(52.02) = 121.7 \text{ kJ}$

$\Rightarrow Q > 0 \Rightarrow$ heat must be added to the reactor

2) We have: mass of sample: $m = 2.7 \text{ g}$

Water mass: $m_w = 650 \text{ g}$

Temperature change: $\Delta T = 36.2 - 23.6 = 12.6^\circ\text{C}$

$C_{\text{water}} = 4.184 \text{ J/g} \cdot ^\circ\text{C}$

$q_{\text{water}} = m_w c \Delta T = (650)(4.184)(12.6) = 34264 \text{ J} = 34.26 \text{ kJ}$

• Bomb calorimeter $\Rightarrow$ constant volume

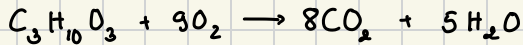
$q_{\text{rxn}} = \Delta U$

• Heat released by reaction = heat gained by water:

$q_{\text{rxn}} = -34.26 \text{ kJ}$

$$\Delta U = \frac{-34.26}{2.7} \Rightarrow \Delta U = -12.69 \text{ kJ/g}$$

Balance Reaction:



$$\Delta H = \Delta U + \Delta n_g RT$$

$$\Delta n_g = 8 - 9 = -1 \quad M = 154 \text{ g/mol}$$

$$\Delta U = (-12.69)(154) = -1955 \text{ kJ/mol}$$

$$\Delta H = -1955 + (-1)(8.314 \times 10^{-3})(298)$$

$$\Rightarrow \Delta H = -1957.5 \text{ kJ/mol}$$

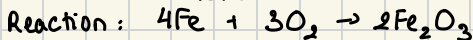
$$\Rightarrow \Delta H = \frac{-1957.5}{154} = -12.71 \text{ kJ/g}$$

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3) We have: Heat release rate: 100 kJ/s

Time: 6 s

$$\Rightarrow Q_{\text{total}} = 100 \times 6 = 600 \text{ kJ}$$



$$\Delta H_f^\circ(\text{Fe}_2\text{O}_3) \approx -824 \text{ kJ/mol}$$

$$\Rightarrow \Delta H = 2(-824) = -1648 \text{ kJ}$$

$\Rightarrow$ This corresponds to 4 mol Fe

$$\text{Heat per mol Fe: } \frac{-1648}{4} = -412 \text{ kJ/mol Fe}$$

$$n_{\text{Fe}} = \frac{600}{412} = 1.46 \text{ mol}$$

$$m_{\text{Fe}} = (1.46)(55.84) = 81.5 \text{ g Fe}$$

From stoichiometry: $4\text{Fe} \rightarrow 3\text{O}_2$

$$\Rightarrow n_{\text{O}_2} = \frac{3}{4}(1.46) = 1.095 \text{ mol}$$

$$\Rightarrow m_{\text{O}_2} = (1.095)(32) = 35 \text{ g}$$

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Initial mass = 81.5 g

Final mass = Fe + O = 81.5 + 35 = 116.5 g

$$\Rightarrow \Delta m = 116.5 - 81.5 = 35 \text{ g}$$

$$\% \Delta m = \frac{35}{81.5} \times 100 = 43\%$$

4) a) Balanced reaction: $\text{N}_2 + \text{O}_2 \rightarrow 2\text{NO}$

$$\Delta H_r^\circ = \sum \nu \Delta H_f^\circ (\text{products}) - \sum \nu \Delta H_f^\circ (\text{reactants})$$

$$\Rightarrow \Delta H_r^\circ = [2 \times \Delta H_f^\circ (\text{NO})] - [\Delta H_f^\circ (\text{N}_2) + \Delta H_f^\circ (\text{O}_2)]$$
$$= [2(90.25)] - (0 + 0)$$

$$\Rightarrow \Delta H_r^\circ = 180.5 \text{ kJ/mol}$$

b) Balanced reaction: $\text{C}_5\text{H}_{12} + \frac{11}{2} \text{O}_2 \rightarrow 5\text{CO} + 6\text{H}_2\text{O}(\ell)$

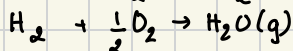
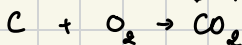
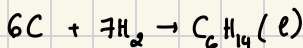
$$\Delta H_r^\circ = \sum \nu \Delta H_f^\circ (\text{products}) - \sum \nu \Delta H_f^\circ (\text{reactants})$$

$$= [5(-110.5) + 6(-285.8)] - [-146]$$

$$\Rightarrow \Delta H_r^\circ = -2121.3 \text{ kJ/mol}$$

c) Balance reaction: $\text{C}_6\text{H}_{14}(\ell) + \frac{19}{2} \text{O}_2 \rightarrow 6\text{CO}_2 + 7\text{H}_2\text{O}(\text{g})$

Formation reactions:



$$\Delta H_r^\circ = \sum \nu \Delta H_f^\circ (\text{products}) - \sum \nu \Delta H_f^\circ (\text{reactants})$$

$$\Rightarrow \Delta H_r^\circ = [6\Delta H_f^\circ (\text{CO}_2) + 7\Delta H_f^\circ (\text{H}_2\text{O})] - [\Delta H_f^\circ (\text{C}_6\text{H}_{14})]$$

$$\Rightarrow \Delta H_r^\circ = [6(-393.5) + 7(-241.8)] - (-198.7)$$

$$\Rightarrow \Delta H_r^\circ = -3854.9 \text{ kJ/mol}$$

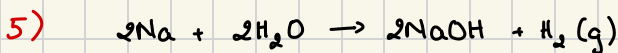
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d) Balanced reaction: $\text{Na}_2\text{SO}_4 + 4\text{CO} \rightarrow \text{Na}_2\text{S} + 4\text{CO}_2$

$$\Delta H_r^\circ = \sum \nu \Delta H_f^\circ(\text{products}) - \sum \nu \Delta H_f^\circ(\text{reactants})$$

$$\Rightarrow \Delta H_r^\circ = [(-364) + 4(-393.5)] - [(-1387) + 4(-110.5)]$$

$$\Rightarrow \Delta H_r^\circ = -109 \text{ kJ/mol}$$



$$1\text{ml H}_2\text{O} \approx 1\text{g} \Rightarrow n_{\text{H}_2\text{O}} = \frac{1}{18} = 0.0556 \text{ mol}$$

$\Rightarrow$ Water is limiting

$$n_{\text{H}_2} = \frac{0.0556}{2} = 0.0278 \text{ mol}$$

$$V_{\text{H}_2} \approx 0.0278 \times 24.5 = 0.68 \text{ L}$$

$$\Delta H \approx -92 \text{ kJ/mol H}_2\text{O}$$

$$\Rightarrow q = (0.0556)(92) = 5.1 \text{ kJ}$$

Jg 5.1 kJ heats only 1g water:

$$\Delta T = \frac{5100}{(1)(4.18)} = 1200^\circ\text{C}$$

$\Rightarrow$ Not realistic physically, but proves:

- + Water instantly vaporizes
- + System becomes extremely hot

Happen: + Sodium melt

- + Water flashes to steam
- + NaOH forms
- + Reaction becomes violent

It is dangerous because:

- + $\sim 0.68 \text{ L H}_2$ gas produced rapidly
- + $\sim 5 \text{ kJ}$ released in a tiny region $\rightarrow$ very high local temperature
- + Temperature high enough to ignite H_2
- $\Rightarrow$ Lead to: Fire, explosion, molten Na splashing, hot NaOH burns